Amazon EC2 and EBS : Takeaway Points

1- Reserved EC2 Instances provide cost savings when you can commit to running instances full time, such as to handle the base traffic.
2- On-Demand EC2 Instances provide the flexibility to jandle traffic spikes, such as on the last day of the month.
3- SPOT EC2 Instances are VERY 'VERY COST-EFFECTIVE' way to address temporary compute needs that are not urgent and are tolerant of interruption.
4- Reserved EC2 Instances are inappropriate for temporary workloads.
5- On-Demand EC2 Instances are good for temporary workloads **BUT** do not offer the cost-savings of Spot EC2 Instances.
6- The Amazon EC2 instance-ID is assigned by AWS as part of the launch-process.
7- For Amazon Windows EC2 instances, the administrator password is assigned by AWS and encrypted via the Public-Key.
8- Remember: The instance type defines the virtual-hardware and the AMI defines the initial software state. You must specify both upon EC2-Launch.
9- You can change the EC2 instance type within the same EC2 instance-family or you can change the Availability Zone (AZ) of your EC2 instance.
10- When there are multiple Security Groups (SGs) associated with an instance, al the rules are aggregated.
11- Enhanced Networking in AWS implies:
    - More Packets Per Second (PPS)
    - Lower Latency
    - Less Jitter (Less unsteadiness)
12- In a HPC (High Performance Cluster of Amazon EC2 instances)we require
    - Low Latency
    - High Bandwidth between the EC2 instances.
    - We can use instance type with 10 Gbps network performance.
    - We can put the instances in a **Placement Group**
    - Enable 'Enhanced Networking on the EC2 instances within our HPC or Placement Group.'
13- Dedicated EC2 instances will NOT SHARE underlying-hosts with other accounts.
14- EC2 Instance Stores : are Low-durability, High-IOPS storage that is included for free (with the hourly cost of an instance.)
15- There are no Tapes in the AWS Infrastructure. Amazon EBS Volumes persist when the instance is stopped.
16- For an Amazon EBS Volume, the data is automatically replicated within an Availability Zone (AZ.)
17- Amazon EBS Volumes can be encrypted upon creation and used by an instance in the same manner as if they were not encrypted.
18- There is NO DELAY in processing when creating a Snapshot.

19- When we’re restoring an Amazon EBS volume from a snapshot, the data on the EBS-volume will be available IMMEDIATELY !